

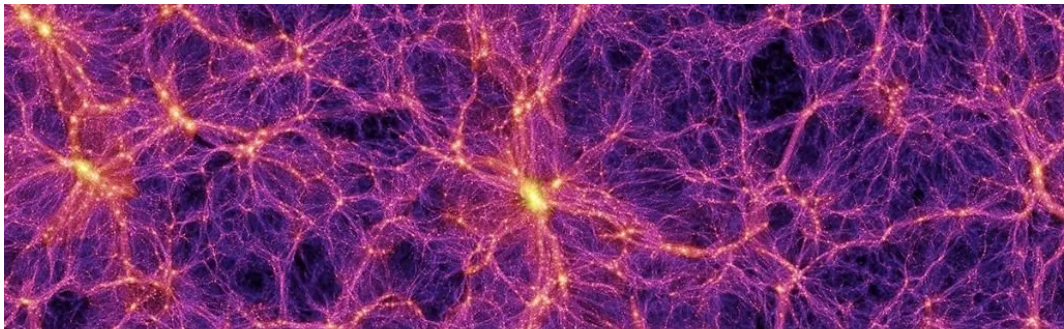
Big Bang Nucleosynthesis (BBN)

Building the first nuclei with a gentle introduction to cosmology

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Outline

How to Mix Physical Cosmology and Buddhism to Lower Everyone's IQ

A gentle introduction to cosmology

The hot Universe: equilibrium & freeze-out

Particle content & thermal history

The cosmic neutrino background

The neutron-to-proton ratio

Nucleosynthesis: building the nuclei

Observations & cosmological significance

Summary

**How to Mix Physical Cosmology and Buddhism to Lower Everyone's
IQ**

A Common Trap: Mixing Two Different Fields

The Misunderstanding

People often compare modern cosmology with ancient Buddhist philosophy, claiming they either prove each other right (or clash with each other)

This usually happens in two ways

- ▶ **Claiming Religion Predicted Science:** "The Buddha predicted the Big Bang or origin of universe!" (Using physics to *validate* ancient ideas)
- ▶ **Claiming Science Replaces Religion:** "Modern science disproves ancient spiritual ideas!" (Using physics to *dismiss* teachings)

Both claims assume that Cosmology (or physics) and Buddhism are trying to solve the exact same problem

Popular Comparisons (And Why They Mismatch)

Buddhist Idea		Physical Science Concept
Paiccasamuppda (Cause & Effect)	↔	The Big Bang / Cosmic Origins
Kalpa (World Cycles)	↔	Cyclic / Multiverse Models
Sansra (Endless Wandering)	↔	Infinite Space-Time
Anicca (Change & Emptiness)	↔	Quantum Fields & Dynamic Particles

A Reality Check

The Uncomfortable Truth

Cosmology and Buddhist thought use **different tools**, pursue **different goals**, and mean **entirely different things** when they talk about "origins."

Key Takeaways for You

- ▶ Comparing Physical Cosmology directly to enlightenment is like measuring mass in meters. It isn't profound, it's just a **category error**
- ▶ Claiming they are "two sides of the same coin" usually just proves you haven't deeply studied either
- ▶ STEM experts aren't experts in ancient Pali suttas, and 5th-century monastics didn't own particle accelerators
- ▶ Metaphors are fine, but "experts" forcing a comparison between incompatible fields just lowers the room's collective IQ.. *best to avoid such debates entirely*

Difference 1: How We Get Knowledge

Physical Cosmology	Buddhist Thought
Telescope data, particle detectors	Textual analysis & logical arguments
Math equations & models	Deductive reasoning
Testing, falsifiability, repeatability	Meditation & inner mental observation
Models updated when new data comes in	Practical guide to personal peace

Neither field has the experimental tools to test or disprove the claims of the other

Difference 2: What is the Main Goal?

Cosmology: Description

- ▶ Explains **how** matter, energy, space, and time behave
- ▶ Neutral: Unconcerned with human emotion, ethics, or personal meaning

Buddhism: Practical Help

- ▶ Helps people reduce mental suffering (*dukkha*)
- ▶ Analyzes nature only to help human minds find peace and freedom

Knowing the equations of dark energy does not solve human anxiety or grief, and achieving peace of mind won't give you a science degree. They are completely different skill sets

Difference 3: What Does "Origin" Mean?

Cosmological "Origin"

- ▶ A physical event in time (~ 13.8 billion years ago).
- ▶ Asks: "What happened first in the history of space-time?"

Buddhist "Origin" (Cause & Effect)

- ▶ A rule happening right now ("When A is present, B happens")
- ▶ Asks *how do mental states depend on conditions in this present moment?*

Confusing an ancient cosmic explosion with a present rule of cause-and-effect is a basic misunderstanding

The Parable of the Poisoned Arrow

The Parable of the Poisoned Arrow

(MN 63)

If a man is shot with a poisoned arrow, he shouldn't waste time asking who shot it, what caste they belonged to, or what bow string was used. He needs to pull the arrow out! Speculating on cosmic origins doesn't solve human suffering.

The Nature of Scientific Models

- ▶ Cosmology relies on clear, testable evidence (Expanding Universe, Cosmic Microwave Background, Light Element Abundances)
- ▶ Scientific models (Λ CDM) are *provisional* and they update whenever new data arrives

Why Linking Spirituality directly to Science is Risky

- ▶ If you tie spiritual truth to today's physics paper, your belief becomes vulnerable when physics updates tomorrow
- ▶ Using astrophysics to "disprove" *kamma* misses the point of why people believe *kamma* in the first place.

Clearing Up a Common Misconception

But isn't the Big Bang just a theory?

In Everyday Speech

a theory = a guess

A hunch or an unproven idea:

I have a theory about who ate the cake.

In Science

a theory = tested explanation

A framework repeatedly confirmed by evidence that makes testable predictions.

Key Takeaway

Gravity, germ theory, and atomic theory are all **theories** too. In science, the word signals a well-established, highly tested explanation, not uncertainty

Summary for Science Students

The Takeaway

Physical cosmology and Buddhist thought serve completely different human needs through completely different methods.

How to Navigate Both Without Losing IQ Points

- ▶ Never treat metaphors from monks or gurus as empirical proof for Cosmology (or Science).
- ▶ Let science describe physical reality (mechanics & data), and let spiritual practice guide mental training (ethics & peace)
- ▶ Your time and IQ are far too valuable to waste on pseudo scientific cross talks

Transitioning from Meta Debate to Hard Science

Philosophical Debate Over: Time for Actual Cosmology!

If someone says *science is just a theory* or tries to substitute quantum mechanics with spiritual teachings:

Tell them to **sit back, shut up, take notes, and watch how science actually works**

- ▶ Now that we've established why blending Buddhism with cosmology is a category error, let's explore the **scientific side of the picture** with rigor
- ▶ We are zooming in on **one specific domain**: how the universe empirically expanded, cooled, and built elements from raw thermodynamics

Welcome to Physical Cosmology. Let's do some actual science!

A gentle introduction to cosmology

What is cosmology?

Cosmology is the scientific study of the Universe as a whole (its origin, evolution, content and fate).

- ▶ It is a peculiar science: can the *container* of all objects be studied as if it were one of the objects inside it?
- ▶ The answer is essentially yes, but at a price: we must assume the **Cosmological Principle**.
→ the Universe is **homogeneous** and **isotropic**, both in *content* and in *laws* (General Relativity holds everywhere).

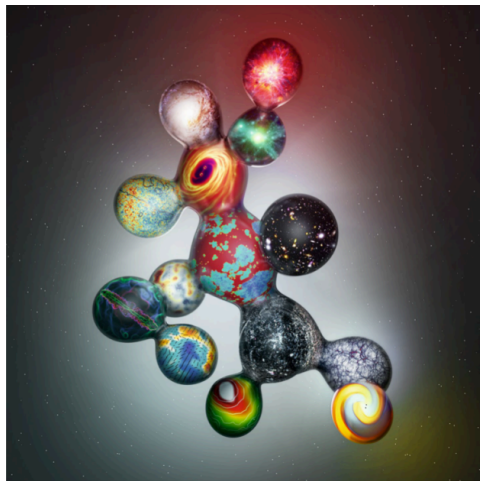
Take away

Cosmology is the science that *tests* the Cosmological Principle. Break it, and cosmology as a predictive science is in trouble.

The Universe is not static: four fundamental facts

1. Distant galaxies are **redshifted** $\Rightarrow$ the Universe *expands*.
2. Matter is $\sim 75\%$ H and $\sim 25\%$ He by mass $\Rightarrow$ the Universe was once *hot and dense*.
3. It is bathed in low-energy photons, the **CMB**, a $T \sim 2.7$ K black body $\Rightarrow$ a hot past in thermal equilibrium.
4. The distant Universe is measurably *hotter* (e.g. $T \sim 5.08$ K at $z = 0.89$).

All four point to the same picture: an **expanding, cooling Universe with a hot beginning**.



A spectral feature emitted at λ_e is observed at $\lambda_o > \lambda_e$. We quantify the shift with the **redshift** z :

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e}.$$

- ▶ $z = 0$: no recession $z > 0$: reddening.
- ▶ The further the galaxy, the larger the z .

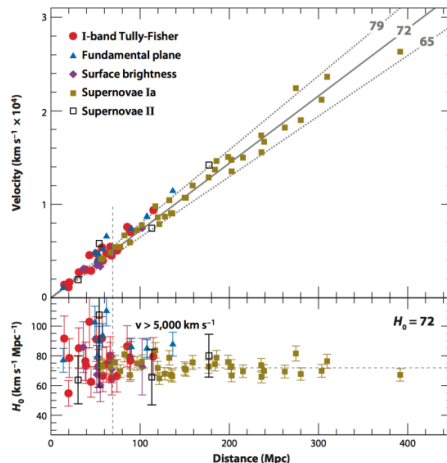
Careful: cosmological redshift is **not** a Doppler effect. The reddening is produced by the *evolution of distances during the light's journey*, not by the motion of the source through space.

Hubble's law

Recession velocity grows linearly with distance:

$$v = H_0 d, \quad H_0 \simeq 70 \text{ km/s/Mpc}.$$

- ▶ A galaxy at 100 Mpc recedes at $\sim 7000 \text{ km/s}$.
- ▶ Planck: $H_0 \simeq 67$; SNe: $H_0 \simeq 72 \Rightarrow$ the “ H_0 tension”.
- ▶ Reminder: $1 \text{ Mpc} \sim 3 \times 10^{22} \text{ m} \sim 3 \text{ million ly}$.



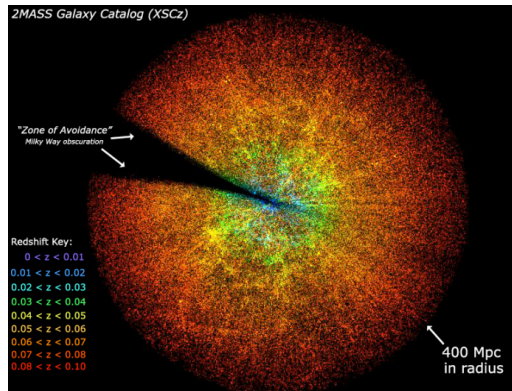
An expanding Universe and the scale factor

Distances between comoving points grow with time through a single **scale factor** $a(t)$:

$$r(t) = a(t) r_0.$$

- ▶ r_0 : **comoving** distance (time-independent).
- ▶ $r(t)$: **physical** / **proper** distance.
- ▶ $a(t)$ is dimensionless, the same for all objects, and $a(t) > 0$.
- ▶ Convention: today $a(t_0) = a_0 = 1$.

$$H(t) = \frac{\dot{a}}{a}, \quad v = \dot{r} = H(t) r(t).$$



Redshift *is* a measurement of the scale factor

For a photon travelling in an expanding FRW universe one finds that wavelengths stretch exactly like all other distances:

$$\lambda_R = \lambda_E \frac{a(t_R)}{a(t_E)}.$$

Received today ($a_R = 1$):

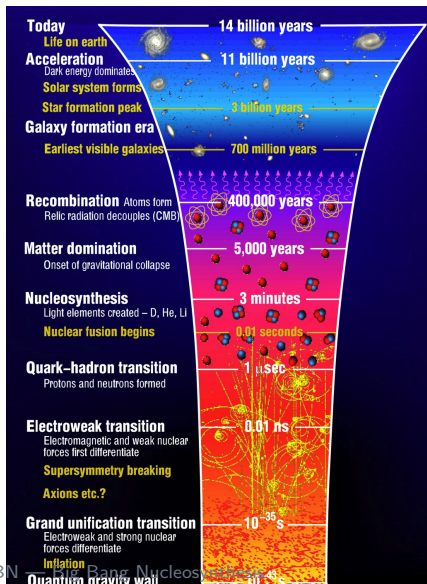
$$\boxed{1 + z = \frac{1}{a_E}} \quad \Longleftrightarrow \quad a = \frac{1}{1 + z}.$$

- ▶ $z = 1 \Rightarrow a = 0.5$: distances were half of today's.
- ▶ $z = 10 \Rightarrow a \simeq 0.09$.
- ▶ Recombination: $z \simeq 1100 \Rightarrow a \simeq 10^{-3}$.

Take away

Measuring z is measuring the size of the Universe when the light was emitted — and therefore a clock reading of cosmic history.

The Universe has a history

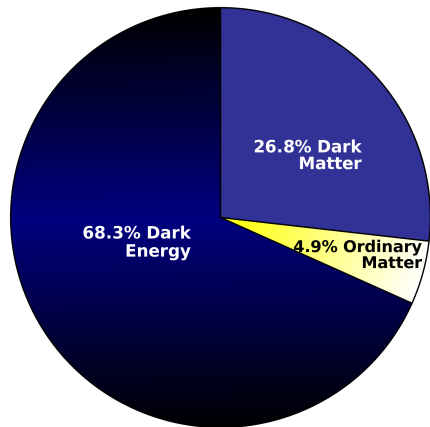


Epoch	Time
Big Bang / Planck time	$\sim 10^{-44}$ s
Inflation	$\sim 10^{-35}$ s
Nucleosynthesis (BBN)	~ 3 min
Recombination / CMB	380 000 yr
First stars (cosmic dawn)	~ 200 Myr
Reionization	~ 1 Gyr
Accelerated expansion	since ~ 7 Gyr
Today	~ 13.8 Gyr

BBN happens very early — in the first few minutes, deep in the radiation-dominated era.

The content of the Universe

The dynamics of $a(t)$ are driven by the **energy content**. Today's budget (Λ CDM, the “concordance model”):



- ▶ Baryons (ordinary matter): $\Omega_b \simeq 0.05$
- ▶ Dark matter: $\Omega_{\text{dm}} \simeq 0.27$
- ▶ Dark energy (vacuum): $\Omega_\Lambda \simeq 0.68$
- ▶ Radiation: $\Omega_r \simeq 10^{-4}$ (tiny *today*)

For BBN, only two components matter: the **radiation** that dominated the early Universe, and the **baryons** whose nuclei we will build. The amount of baryons is set by a single number η .

The Friedmann equation

General Relativity, applied to a homogeneous & isotropic fluid, links the acceleration of a to energy density ε and pressure P :

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2} (\varepsilon + 3P), \quad \varepsilon = \rho c^2.$$

Integrating (and summing over fluids) gives the expansion rate:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = H_0^2 \left(\frac{\Omega_m}{a^3} + \frac{\Omega_r}{a^4} + \Omega_v - \frac{\Omega_K}{a^2} \right)$$

Each fluid dilutes differently with a :

$$\varepsilon_m \propto a^{-3}, \quad \varepsilon_r \propto a^{-4}, \quad \varepsilon_v = \text{const.}$$

Domination eras and the early Universe

Because the three fluids scale differently, the Universe passes through three regimes:

$$\underbrace{a \propto t^{1/2}}_{\text{radiation}} \longrightarrow \underbrace{a \propto t^{2/3}}_{\text{matter}} \longrightarrow \underbrace{a \propto e^{H_0 t}}_{\text{vacuum}}$$

- ▶ Radiation ($\propto a^{-4}$) always wins at the *smallest* $a \Rightarrow$ earliest times.
- ▶ BBN takes place during radiation domination, so throughout this lecture:

$$a(t) \propto \sqrt{t}$$

This single relation controls how fast the young Universe expands and cools.

Temperature and the scale factor

Radiation (photons) redshift as the Universe expands. Their number is conserved and diluted only geometrically ($n_\gamma \propto a^{-3}$), so the black-body temperature obeys:

$$\boxed{T = \frac{T_0}{a} = T_0(1+z)}, \quad T_0 = 2.73 \text{ K}.$$

Combined with $a \propto \sqrt{t}$ during radiation domination, temperature is a clock:

$$T \sim \frac{10^{10} \text{ K}}{\sqrt{t(\text{s})}} \quad \Longleftrightarrow \quad k_B T \sim \frac{1 \text{ MeV}}{\sqrt{t(\text{s})}}.$$

Take away

$t \sim 1 \text{ s} \Leftrightarrow k_B T \sim 1 \text{ MeV}$. This is exactly the energy scale of nuclear & weak physics — the reason BBN happens when it does.

Setting the stage for BBN

In the first minutes the Universe is:

- ▶ **hot** ($k_B T \sim \text{MeV}$, $T \sim 10^{9-10} \text{ K}$),
- ▶ **dense**,
- ▶ **radiation-dominated** ($a \propto \sqrt{t}$),
- ▶ in near-perfect **thermal equilibrium** (reactions are fast).

Under these conditions nuclear fusion *can* occur. The question of BBN is:

which nuclei form, and in what proportions?

The hot Universe: equilibrium & freeze-out

The central quantity: abundances

We track **number densities** $n(t)$ (particles per m^3) of each species. Two things change an abundance:

1. **Cosmological dilution** — expansion spreads particles out.
2. **Reactions** — creation & destruction of particles.

The whole story of BBN is a *competition* between these two effects. When reactions win, species sit at equilibrium; when expansion wins, they **freeze out**.

Cosmological dilution

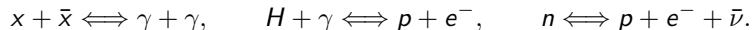
A perfectly passive species (no reactions) still dilutes with the volume:

$$n(a) = \frac{n_0}{a^3} \quad \implies \quad \frac{dn}{dt} = -3 \frac{\dot{a}}{a} n = -3H n.$$

- ▶ The source term is negative: abundance decreases.
- ▶ The characteristic timescale is the **Hubble time** $t_H = H^{-1}$ — the age of the Universe.
- ▶ So pure dilution is *slow*: it acts on cosmic timescales.

Reactions and reaction rates

Abundances also change through reactions, e.g.



For one species, $\frac{dn}{dt} = -\Gamma n$, where the reaction rate is

$$\Gamma \sim v \sigma n_t$$

with v the particle velocity, σ the cross-section, n_t the target abundance. Typical timescale: Γ^{-1} .

Equilibrium as a restoring force

Because the early Universe is hot and dense, reactions are efficient and the abundance stays close to its **equilibrium value** n_e :

$$\frac{dn}{dt} = -\Gamma (n - n_e).$$

- ▶ If $n > n_e$: the abundance is driven *down*.
- ▶ If $n < n_e$: the abundance is driven *up*.

Γ acts like a **callback force** pulling n towards n_e . If we know $n_e(t)$, we can predict the evolution without the full reaction network.

The master equation

Putting dilution and reactions together:

$$\frac{dn}{dt} = -3Hn - \Gamma(n - n_e)$$

Two timescales compete:

- ▶ $\Gamma \gg H$: reactions win $\Rightarrow n \simeq n_e$ (equilibrium, possibly time-dependent through T).
- ▶ $\Gamma \ll H$: expansion wins $\Rightarrow$ only dilution, $n \propto a^{-3}$, and the total number in a comoving volume is conserved: $N = nV \simeq \text{const.}$ The species is **frozen**.

Freeze-out: how H and Γ scale

During radiation domination $a \propto \sqrt{t}$, so

$$H = \frac{\dot{a}}{a} \sim t^{-1} \sim a^{-2}.$$

Meanwhile $\Gamma \sim v\sigma n_t$ with σ const, $n_t \propto a^{-3}$ and $v \propto \sqrt{T} \propto a^{-1/2}$:

$$\Gamma \sim a^{-3.5}.$$

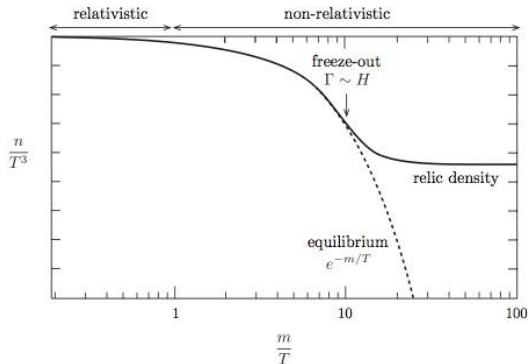
	early (a small)	late (a large)
Rates	$\Gamma > H$	$\Gamma < H$
State	equilibrium	frozen

Γ falls faster than H : every species eventually freezes out. The crossover $\Gamma \simeq H$ defines the **freeze-out**.

Freeze-out: the picture

- ▶ Early: $\Gamma \gg H$ — equilibrium follows $n_e(T)$.
- ▶ Crossover: $\Gamma \sim H$ — the “freeze-out”.
- ▶ Late: $\Gamma \ll H$ — abundance frozen, only diluting.

The *frozen* value is a **relic** of the conditions at freeze-out. BBN is a sequence of such freeze-outs (neutrinos, the n/p ratio, the nuclei themselves).



Equilibrium abundances: statistical mechanics

Particles are treated as free, with energy set by momentum:

$$E^2 = m^2 c^4 + p^2 c^2.$$

In a box of size L , momentum states are quantised, $\vec{p} = \hbar \vec{k}$, $\vec{k} = \frac{2\pi}{L}(n_x, n_y, n_z)$. The equilibrium abundance sums over all states (Grand-canonical ensemble):

$$n = \frac{g}{(2\pi\hbar)^3} \int_{p_{\min}}^{\infty} \frac{4\pi p^2 dp}{e^{(E(p)-\mu)/k_B T} \pm 1}.$$

- ▶ +1: fermions (Fermi–Dirac); –1: bosons (Bose–Einstein).
- ▶ g : degeneracy, μ : chemical potential.

Photon abundance

For photons $g = 2$, $\mu = 0$, $p_{\min} \simeq 0 \Rightarrow$ the black-body integral:

$$n_\gamma = \frac{8\pi}{(2\pi\hbar)^3} \int_0^\infty \frac{p^2 dp}{e^{pc/k_B T} - 1} \simeq 0.244 \left(\frac{k_B T}{\hbar c} \right)^3.$$

With $T = 2.7$ K today:

$$n_\gamma \simeq 413 \text{ photons/cm}^3$$

Photons dominate the *number* budget of the Universe by a huge margin — a fact that turns out to be crucial for BBN.

The baryon-to-photon ratio η

The most important single number for BBN. Comparing baryons to photons:

$$\eta = \frac{n_b}{n_\gamma} \sim 5 \times 10^{-10} \left(\frac{\Omega_b h^2}{0.02} \right).$$

- ▶ There are $\sim 10^9$ photons for every baryon.
- ▶ η is constant (both $n_b, n_\gamma \propto a^{-3}$ once frozen).
- ▶ Its smallness will **delay** nucleosynthesis and set the final abundances.

Take away

η measures how much ordinary matter the Universe contains. BBN and the CMB both measure it — and they agree.

The temperature–time relation (revisited)

Since $N_\gamma \simeq \text{const}$, $n_\gamma \propto a^{-3} \propto T^3$, giving

$$T = \frac{T_0}{a} = T_0(1+z).$$

During radiation domination ($a \propto \sqrt{t}$):

$$T \sim \frac{1}{\sqrt{t}} \quad \Rightarrow \quad T \sim \frac{10^{10} \text{ K}}{\sqrt{t(\text{s})}} \sim \frac{1 \text{ MeV}}{\sqrt{t(\text{s})}}.$$

This converts any energy threshold into a cosmic epoch: a reaction with a characteristic energy $k_B T^*$ “switches” at a specific time.

Ultra-relativistic vs non-relativistic species

The mass m sets a critical temperature $k_B T \sim mc^2$:

Ultra-relativistic (UR): $k_B T \gg mc^2$

$$E \simeq pc, \quad n_e \sim n_\gamma$$

Abundant, comparable to photons.

Non-relativistic (NR): $k_B T \ll mc^2$

$$E \simeq mc^2 + \frac{p^2}{2m}$$

$$n_e \sim (k_B T)^{3/2} e^{-mc^2/k_B T}$$

Boltzmann-suppressed — the species *disappears*.

Since T falls with time, every massive species eventually crosses from UR to NR. Heavy particles do so early, light particles late.

Relic particles

The generic life story of a massive species:

$$\underbrace{n_e \sim n_\gamma}_{\text{UR, abundant}} \longrightarrow \underbrace{n_e \sim e^{-mc^2/k_B T}}_{\text{NR, annihilating away}} \longrightarrow \underbrace{n = \frac{n_0}{a^3}}_{\text{frozen relic}}$$

- ▶ Annihilation would erase the species — *but* freeze-out halts it, leaving a **relic** abundance.
- ▶ Particles that freeze out *while still UR* (e.g. neutrinos) stay very abundant $\Rightarrow$ cosmic backgrounds.

This freeze-out logic is exactly what we now apply to neutrinos, then to neutrons and protons.

Particle content & thermal history

The Standard Model of particle physics (reminder)

- ▶ **Quarks** (6): fermions, fractional charge; feel strong, weak & EM forces. Build nucleons:
 $p = uud (+1)$, $n = udd (0)$.
- ▶ **Leptons** (6): e, μ, τ and their neutrinos ν_e, ν_μ, ν_τ ; feel weak & EM (neutrinos: weak only).
- ▶ **Bosons**: force carriers — γ (EM), $W^\pm, Z$ (weak), gluons g (strong). Mass from the Higgs field.

Q U A R K S	UP mass 2,3 MeV/c ² charge $\frac{2}{3}$ spin $\frac{1}{2}$ 	CHARM 1,275 GeV/c ² $\frac{2}{3}$ $\frac{1}{2}$ 	TOP 173,07 GeV/c ² $\frac{2}{3}$ $\frac{1}{2}$ 	GLUON 0 0 1 	HIGGS BOSON 126 GeV/c ² 0 0 0 
	DOWN 4,8 MeV/c ² $-\frac{1}{3}$ $\frac{1}{2}$ 	STRANGE 95 MeV/c ² $-\frac{1}{3}$ $\frac{1}{2}$ 	BOTTOM 4,18 GeV/c ² $-\frac{1}{3}$ $\frac{1}{2}$ 	PHOTON 0 0 1 	G A U G E B O S O N S
	ELECTRON 0,511 MeV/c ² -1 $\frac{1}{2}$ 	MUON 105,7 MeV/c ² -1 $\frac{1}{2}$ 	TAU 1,777 GeV/c ² -1 $\frac{1}{2}$ 	Z BOSON 91,2 GeV/c ² 0 1 	
	ELECTRON NEUTRINO <2,2 eV/c ² 0 $\frac{1}{2}$ 	MUON NEUTRINO <0,17 MeV/c ² 0 $\frac{1}{2}$ 	TAU NEUTRINO <15,5 MeV/c ² 0 $\frac{1}{2}$ 	W BOSON 80,4 GeV/c ² ±1 1 	

Baryons — what BBN builds

- ▶ **Baryons** = protons, neutrons (and the nuclei they form). A shorthand for “ordinary / visible matter”.
- ▶ They interact with photons — which is why they were tightly coupled to radiation in the early Universe.
- ▶ BBN is the story of how the free protons and neutrons of the first second assemble into the *first nuclei*.

Take away

Everything downstream depends on two frozen numbers: the **baryon-to-photon ratio** η and the **neutron-to-proton ratio** n/p .

A thermal history of the early Universe

T	$k_B T$	t	Event
$3 \times 10^{12} \text{ K}$	250 MeV	10^{-5} s	quarks confined in nucleons
10^{12} K	90 MeV	—	pions, μ , ν_μ disappear
$\lesssim 10^{11} \text{ K}$	9 MeV	—	n/p asymmetry sets in
$6 \times 10^9 \text{ K}$	500 keV	$\sim 1 \text{ s}$	$e^\pm$ become NR; ν_e freeze-out
10^9 K	—	$\sim 100 \text{ s}$	nucleosynthesis starts
3000 K	—	380 000 yr	recombination, CMB released

BBN sits in the window $\sim 1 \text{ s} \rightarrow \sim 3 \text{ min}$: from neutrino/ np freeze-out to the end of light-element fusion.

The three characters of the BBN story

Everything that follows is three freeze-outs in sequence:

1. **Neutrinos** decouple ($t \sim 1 \text{ s}$) $\Rightarrow$ cosmic neutrino background.
2. **The n/p ratio** freezes ($t \sim 1 \text{ s}$) $\Rightarrow$ sets how many neutrons are available.
3. **Nuclei** form once the “deuterium bottleneck” breaks ($t \sim 3 \text{ min}$) $\Rightarrow$ locks in the light-element abundances.

We take them one at a time.

Key milestone: $k_B T \sim 1 \text{ MeV}$, $t \sim 1 \text{ s}$

At $t \sim 1 \text{ s}$ several things happen *at once*:

- ▶ e^+e^- start to become non-relativistic.
- ▶ Electron neutrinos ν_e freeze out.
- ▶ Weak reactions interconverting $n \leftrightarrow p$ freeze.

This single epoch fixes two of our three characters — the neutrino background and the n/p ratio.

The cosmic neutrino background

Neutrino coupling and decoupling

Before $t \sim 1\text{ s}$, ν_e , $e^\pm$ and γ are all coupled through weak and electromagnetic interactions, sharing one temperature:

$$T_\nu = T_\gamma.$$

As $e^\pm$ turn non-relativistic and annihilate, the electron abundance drops, and the weak reactions that kept neutrinos in the bath switch off:

$$\Gamma_{\text{weak}} < H \quad \Rightarrow \quad \nu_e \text{ freeze out at } t \sim 1\text{ s}.$$

Neutrinos are far lighter than 1 MeV, so they freeze out *while still relativistic* $\Rightarrow$ they stay abundant.

e^+e^- annihilation heats the photons

After neutrinos decouple, e^+e^- pairs annihilate:

$$e^+ + e^- \rightarrow \gamma + \gamma.$$

- ▶ Neutrinos are already *gone from the bath*: their temperature is essentially unchanged, $T'_\nu = T_\nu$.
- ▶ Photons *do* still couple to $e^\pm$: they inherit the entropy of the annihilating pairs — effectively, new light is created.

Result: photons get a relative “reheat” compared to neutrinos.

The $(4/11)^{1/3}$ factor

Entropy conservation before/after annihilation. In the UR regime bosons and fermions carry comparable entropy, $s_F = \frac{7}{8} \frac{g_F}{g_B} s_B$. With $g_\gamma = g_e = 2$, photons inherit:

$$s'_\gamma = \left(\frac{11}{4}\right) s_\gamma.$$

Since $s \propto T^3$ for a photon gas:

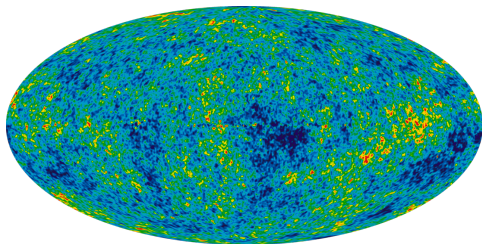
$$T'_\gamma = \left(\frac{11}{4}\right)^{1/3} T_\nu \quad \Longleftrightarrow \quad T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma$$

The relic neutrino background

The neutrino background is slightly *colder* than the CMB:

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma \simeq 1.95 \text{ K} \quad (\text{vs } 2.73 \text{ K}).$$

- ▶ Predicted, but not yet directly detected.
- ▶ A genuine relic from $t \sim 1 \text{ s}$ — even earlier than the CMB.
- ▶ Its density feeds back into the expansion rate H during BBN (the effective number of neutrino species N_{eff}).



Why the neutrino background matters for BBN

The neutrinos are not just spectators:

- ▶ They set part of the radiation energy density, hence H .
- ▶ A faster expansion (more relativistic species) $\Rightarrow$ earlier n/p freeze-out $\Rightarrow$ *more* neutrons $\Rightarrow$ *more* helium.
- ▶ BBN therefore **counts** light species: it constrains $N_{\text{eff}} \simeq 3$, one of its landmark achievements.

Take away

The abundance of helium is sensitive to the number of neutrino families — cosmology as a particle-physics laboratory.

The neutron-to-proton ratio

Initial conditions: free protons and neutrons

Around $t \sim 1$ s, protons and neutrons are already non-relativistic ($mc^2 \sim 1$ GeV vs $k_B T \sim 1$ MeV). They interconvert through weak reactions:

$$n + \nu_e \leftrightarrow p + e^-, \quad n + e^+ \leftrightarrow p + \bar{\nu}_e, \quad n \leftrightarrow p + e^- + \bar{\nu}_e.$$

While these are fast, the ratio follows equilibrium.

n/p in equilibrium

For two NR species differing only by mass, the equilibrium ratio is a Boltzmann factor in the mass difference:

$$\frac{n}{p} \simeq \exp\left(-\frac{(m_n - m_p)c^2}{k_B T}\right), \quad (m_n - m_p)c^2 \simeq 1.29 \text{ MeV}.$$

- ▶ At very high T : $n/p \rightarrow 1$ (equal numbers).
- ▶ As T drops below $\sim 1 \text{ MeV}$, neutrons (heavier) become rarer.

If the reactions stayed fast forever, $n/p \rightarrow 0$ and *no* neutrons would survive. But they don't stay fast.

Weak freeze-out of the n/p ratio

The weak rates fall (like all reaction rates) faster than H . They freeze out at $k_B T \sim 0.8 \text{ MeV}$, $t \sim 1 \text{ s}$, locking in:

$$\left. \frac{n}{p} \right|_{\text{freeze}} \simeq \frac{1}{6}$$

- ▶ Below this T , the equilibrium value is abandoned.
- ▶ The neutron supply is now (almost) fixed — these are the neutrons available to make nuclei.
- ▶ Earlier freeze-out (faster H) would mean *more* neutrons.

β -decay drift before fusion

Free neutrons are unstable ($\tau_n \simeq 880 \text{ s} \approx 15 \text{ min}$). Between freeze-out ($t \sim 1 \text{ s}$) and the onset of fusion ($t \sim 3 \text{ min}$), some neutrons β -decay:

$$n \rightarrow p + e^- + \bar{\nu}_e.$$

So the ratio drifts down from 1/6 to

$$\frac{n}{p} \simeq \frac{1}{7} \text{ at the start of nucleosynthesis.}$$

Timing matters: if fusion started later, even fewer neutrons would remain. The neutron lifetime is a key BBN input.

Why the n/p ratio decides the helium yield

Essentially *all* available neutrons end up locked in helium-4 (two protons + two neutrons). So the helium mass fraction is set almost entirely by n/p :

$$Y_p = \frac{2(n/p)}{1 + (n/p)}.$$

With $n/p \simeq 1/7$:

$$Y_p = \frac{2/7}{8/7} = \frac{1}{4} = 0.25.$$

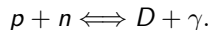
Take away

The famous *25% helium by mass* is, at heart, just the frozen neutron-to-proton ratio $\sim 1/7$.

Nucleosynthesis: building the nuclei

Deuterium: the gateway

Every path to helium goes through **deuterium** (D : 1 proton + 1 neutron), the simplest composite nucleus:



- ▶ You cannot fuse two protons or two neutrons into anything stable — D is the mandatory first step.
- ▶ Binding energy $B_D = (m_p + m_n - m_D)c^2 \simeq 2.2 \text{ MeV}$.
- ▶ No deuterium $\Rightarrow$ no helium. So the whole timing of BBN hinges on when D can survive.

Deuterium equilibrium abundance

Equating chemical potentials $\mu_p + \mu_n = \mu_D$ (with $\mu_\gamma = 0$) and using the NR abundance formula gives, for the ratio $X_D = n_D/n_p$:

$$X_D \sim \eta (k_B T)^{3/2} e^{B_D/k_B T}.$$

- ▶ The Boltzmann factor $e^{B_D/k_B T}$ *favours* deuterium once $k_B T < B_D = 2.2 \text{ MeV}$.
- ▶ But there is a spoiler out front: the tiny prefactor η .

The deuterium bottleneck

Naively, with $B_D = 2.2 \text{ MeV}$ and $k_B T \sim 1 \text{ MeV}$ at $t \sim 1 \text{ s}$, deuterium should form immediately. It does *not*, because $\eta \sim 10^{-10}$:

$$X_D \sim \underbrace{10^{-10}}_{\eta} \times e^{B_D/k_B T}.$$

Deuterium only becomes abundant once the exponential overwhelms η — which requires $k_B T$ to drop well below B_D , to $\sim 0.07 \text{ MeV}$:

$t \sim 3 \text{ minutes, not } 1 \text{ second.}$

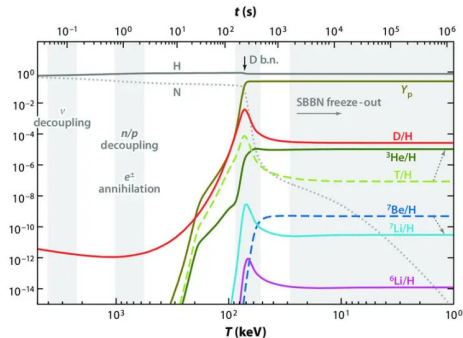
This delay is the **deuterium bottleneck**.

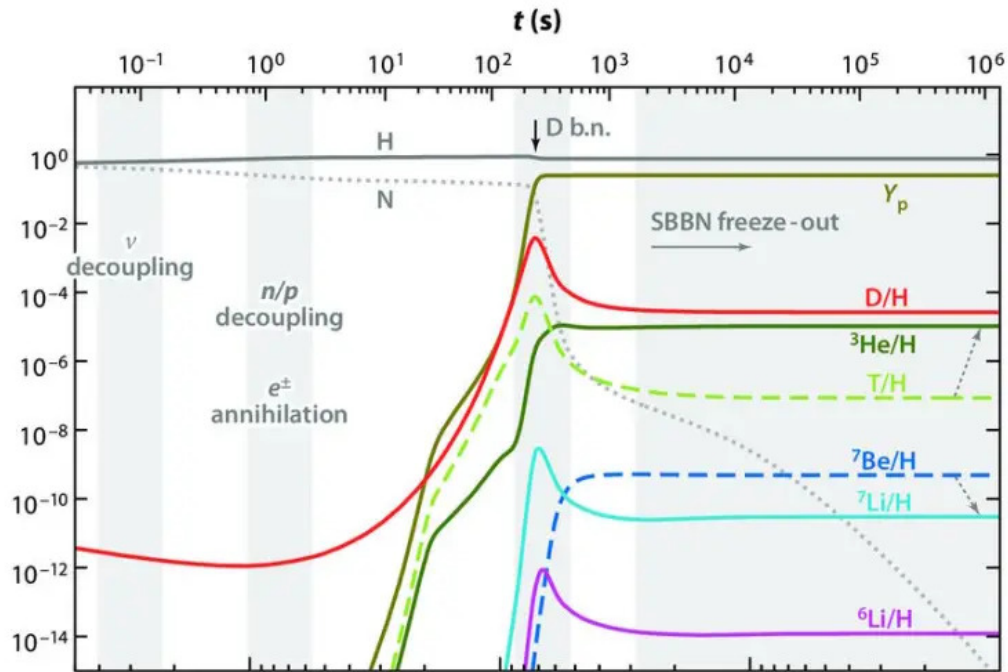
Physical picture of the bottleneck

Why the delay, physically?

- ▶ There are $\sim 10^9$ photons per baryon.
- ▶ Even when $k_B T < B_D$, the *tail* of the photon distribution still contains plenty of photons with $E > 2.2$ MeV.
- ▶ These energetic photons *photo-dissociate* newly formed deuterons faster than they can accumulate.

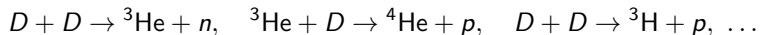
Only when the whole photon bath has cooled enough does D survive — and the floodgates open.





Helium-4 synthesis

Once deuterium survives, fusion proceeds rapidly (helium is very tightly bound), through chains such as:

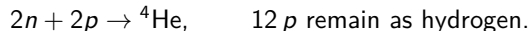


- ▶ Almost all neutrons are swept into ${}^4\text{He}$.
- ▶ By the time this starts, β -decay has left $n/p \simeq 1/7$.

Counting: for every 1 neutron there are 7 protons. One ${}^4\text{He}$ ($2p + 2n$) uses 2 neutrons + 2 protons; a second neutron is needed... net result on the next slide.

The 25% helium mass fraction

With $n : p = 1 : 7$, take 2 neutrons and 2 protons to build one ${}^4\text{He}$; that leaves 12 protons free:



- ▶ By **number**: 1 He nucleus per 12 H nuclei ($\sim 1/12$).
- ▶ By **mass**: He mass $\simeq 4$, total $\simeq 16 \Rightarrow Y_p = 4/16 = 0.25$.

$\sim 75\% \text{ H}, \quad \sim 25\% \text{ He, by mass.}$

This matches the observed cosmic abundances — a triumph of the hot Big Bang.

Why nucleosynthesis stops at helium

BBN essentially halts after ${}^4\text{He}$. Two reasons:

- ▶ **Nuclear structure:** there are *no stable nuclei* at mass number $A = 5$ or $A = 8$. The easy “add a nucleon” ladder is broken right above helium.
- ▶ **Cosmic conditions:** to jump the gap (e.g. triple- α to carbon) needs high density *and* time. A few minutes after the Big Bang, the Universe is already too cold and too dilute.

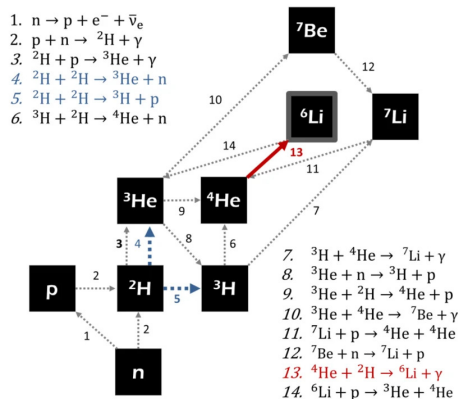
Heavier elements (C, O, Fe, ...) must wait billions of years — they are forged later, **in stars**.

The other products: D, ^3He , ^7Li

BBN does not burn everything into ^4He . Trace amounts survive:

- ▶ **Deuterium** $D/H \sim 10^{-5}$ — leftover from the bottleneck.
- ▶ **Helium-3** $^3\text{He}/H \sim 10^{-5}$.
- ▶ **Lithium-7** $^7\text{Li}/H \sim 10^{-10}$ — a tiny yield.

Each abundance depends on η — so measuring them *measures* the baryon content of the Universe.

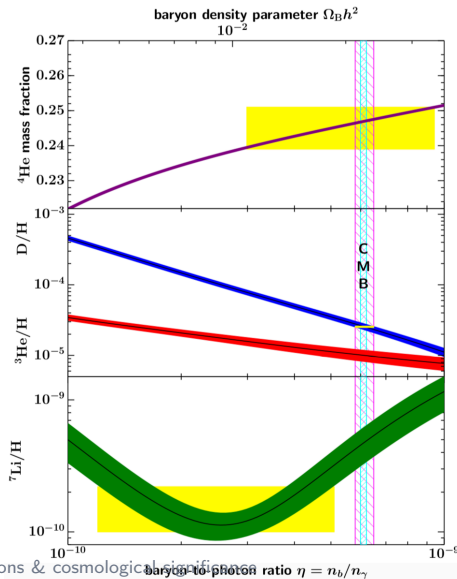


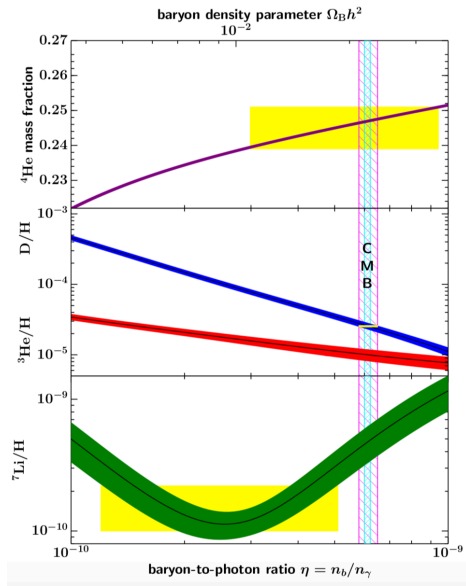
Observations & cosmological significance

Predicted abundances vs the baryon density

The BBN code takes essentially *one* free parameter, η (or $\Omega_b h^2$), and predicts *all* the light-element abundances.

- ▶ Y_p (helium): weak dependence on η .
- ▶ D/H : steep, monotonic — an excellent “baryometer”.
- ▶ ${}^7\text{Li}$: non-monotonic “valley”.





Measuring the primordial helium fraction

^4He is measured in metal-poor extragalactic H II regions, then extrapolated to zero metallicity (zero stellar processing):

$$Y_p^{\text{obs}} \simeq 0.245.$$

In striking agreement with the naive 1/7 neutron-to-proton estimate.

Deuterium: the baryometer

Deuterium is fragile — stars only *destroy* it — so any D we see is a *lower bound* on the primordial value, and high- z gas gives it cleanly:

$$D/H \simeq 2.5 \times 10^{-5}.$$

Because D/H depends steeply on η , this pins down the cosmic baryon density to a few percent.

Two independent measurements of η agree

A cornerstone result of modern cosmology:

Method	epoch probed
BBN (D/H, Y_p)	$t \sim 3 \text{ min}$
CMB acoustic peaks (Planck)	$t \sim 380\,000 \text{ yr}$

Both give $\Omega_b h^2 \simeq 0.022$ — from physics separated by $\sim 400\,000$ years and completely different observables.

Take away

Independent clocks, one answer. This concordance is a pillar of the hot Big Bang model.

The lithium problem

Not everything fits perfectly:

- ▶ BBN + the CMB value of η predict a ${}^7\text{Li}$ abundance $\sim 3\times$ *higher* than measured in old halo stars.
- ▶ Possible culprits: stellar depletion of Li, uncertain nuclear rates, or new physics.

An open, actively debated question — BBN's one persistent blemish.

BBN as a pillar of the hot Big Bang

Alongside the Hubble expansion and the CMB, BBN is one of the three classic observational pillars of Big Bang cosmology. It is remarkable because:

- ▶ It probes the Universe at $t \sim 1\text{--}180\text{ s}$ — the earliest epoch we can test with ordinary nuclear physics.
- ▶ It uses only known, laboratory-measured microphysics.
- ▶ It successfully predicts the abundances of the lightest elements over *ten orders of magnitude* (H to ${}^7\text{Li}$).
- ▶ It doubles as a particle-physics probe (N_{eff} , neutron lifetime, exotic relics).

Summary

The BBN timeline in one slide

Time	$k_B T$	What happens
$\sim 1\text{ s}$	$\sim 1\text{ MeV}$	ν decouple; n/p freezes at $1/6$
$1\text{ s} \rightarrow 3\text{ min}$	$\text{MeV} \rightarrow 0.1\text{ MeV}$	neutrons β -decay: $n/p \rightarrow 1/7$
$\sim 3\text{ min}$	$\sim 0.07\text{ MeV}$	deuterium bottleneck breaks
$\sim 3\text{--}20\text{ min}$	—	rapid fusion $\rightarrow {}^4\text{He}$ (+ traces of D, ${}^3\text{He}$, ${}^7\text{Li}$)
end	—	$\sim 75\%$ H, $\sim 25\%$ He by mass; frozen abundances

Key results to remember

- ▶ The early Universe: hot, dense, radiation-dominated, $a \propto \sqrt{t}$, $k_B T \sim 1 \text{ MeV} / \sqrt{t(\text{s})}$.
- ▶ Everything is governed by **freeze-out**: Γ falls faster than H .
- ▶ Two frozen numbers drive BBN: $\eta \sim 5 \times 10^{-10}$ and $n/p \simeq 1/7$.
- ▶ Helium: $Y_p = \frac{2(n/p)}{1 + n/p} \simeq 0.25$.
- ▶ The tiny η causes the **deuterium bottleneck**, delaying fusion to $\sim 3 \text{ min}$.
- ▶ Abundances agree with observation *and* with the CMB value of η .

Open questions & outlook

- ▶ **Baryogenesis:** why $\eta \sim 10^{-10}$ at all? For a normal freeze-out we would expect $n_b/n_\gamma \sim 10^{-18}$. A matter–antimatter asymmetry (Sakharov conditions) is required, and its origin is unknown.
- ▶ **The lithium problem:** a persistent $\sim 3\sigma$ mismatch.
- ▶ **The cosmic neutrino background:** predicted at 1.95 K, still undetected directly.
- ▶ **New physics:** BBN constrains extra relativistic species, decaying particles, varying constants, ...

BBN: the deepest routinely-testable window on the early Universe.

References & further reading

- ▶ S. Weinberg, *Cosmology*, Oxford (2008).
- ▶ E. Kolb & M. Turner, *The Early Universe*, Addison-Wesley.
- ▶ J. Peacock, *Cosmological Physics*, Cambridge.
- ▶ Mo, van den Bosch & White, *Galaxy Formation and Evolution*.
- ▶ Particle Data Group, *Review of Particle Physics* — “Big-Bang Nucleosynthesis” chapter (updated yearly): pdg.lbl.gov.
- ▶ IPAC / NED Level 5 knowledge base: ned.ipac.caltech.edu/level5/.
- ▶ Course notes: D. Aubert, *Cosmology 3 & 4* (hot Universe / primordial nucleosynthesis).

Thank you — questions?